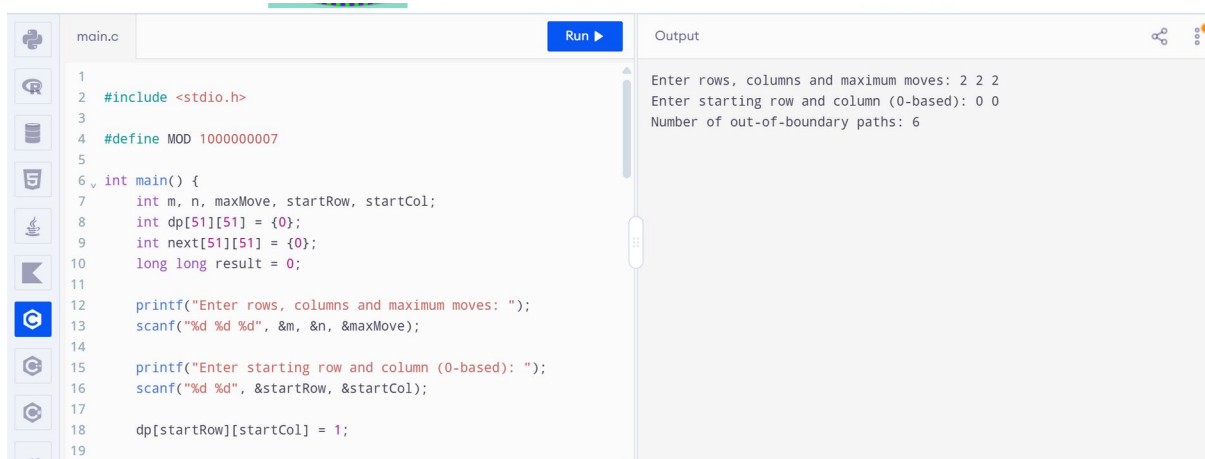


Aim

To find the number of paths by which a ball can move out of the boundary of a grid within a given maximum number of moves using Dynamic Programming.

Algorithm

1. Read the number of rows, columns, maximum moves, and starting position.
2. Initialize a DP table with the number of ways to reach the starting cell as 1.
3. For every move, consider moving up, down, left, and right.
4. If a move goes outside the grid, add the number of ways to the answer.
5. Otherwise, update the number of ways to reach the adjacent cell.
6. Repeat until the maximum number of moves is reached.
7. Display the total number of paths that go outside the grid.



The screenshot shows a C++ IDE with a file named 'main.c'. The code defines a constant MOD as 1000000007 and implements a main function that reads grid dimensions (m, n), maximum moves (maxMove), and starting position (startRow, startCol). It initializes a DP table 'dp' and a 'next' table, both of size 51x51. The program prompts the user for input and calculates the number of paths that exit the grid boundary within the given moves. The output shows the input '2 2 2' for rows, columns, and moves, and '0 0' for starting row and column, resulting in 6 out-of-boundary paths.

```
1
2 #include <stdio.h>
3
4 #define MOD 1000000007
5
6 int main() {
7     int m, n, maxMove, startRow, startCol;
8     int dp[51][51] = {0};
9     int next[51][51] = {0};
10    long long result = 0;
11
12    printf("Enter rows, columns and maximum moves: ");
13    scanf("%d %d %d", &m, &n, &maxMove);
14
15    printf("Enter starting row and column (0-based): ");
16    scanf("%d %d", &startRow, &startCol);
17
18    dp[startRow][startCol] = 1;
19
```

Output:

```
Enter rows, columns and maximum moves: 2 2 2
Enter starting row and column (0-based): 0 0
Number of out-of-boundary paths: 6
```

Complexity Analysis

- Time Complexity: $O(K \times M \times N)$, where K is the maximum number of moves.
- Space Complexity: $O(M \times N)$, because two grid tables are used.